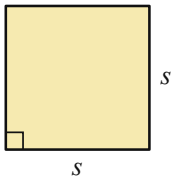
GEOMETRY REFERENCE · PERIMETER, AREA & VOLUME

Formula Sheet

Perimeter is the distance around a flat figure, in units (cm). **Area** is the space inside it, in square units (cm²). **Volume** is the space inside a solid, in cubic units (cm³).



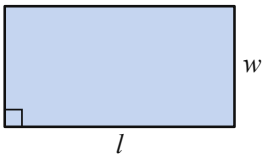
Square

PERIMETER

$P = 4s$

AREA

$A = s^2$



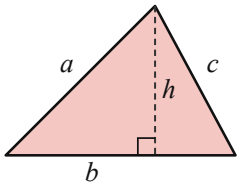
Rectangle

PERIMETER

$P = 2l + 2w$

AREA

$A = lw$



Triangle

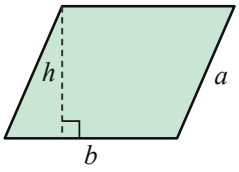
PERIMETER

$P = a + b + c$

AREA

$A = \frac{1}{2}bh$

The height h meets the base b at a right angle.



Parallelogram

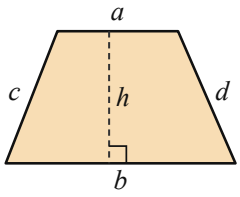
PERIMETER

$P = 2a + 2b$

AREA

$A = bh$

Use the height h, not the slanted side a.



Trapezoid

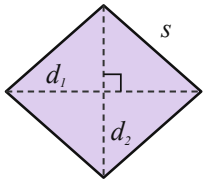
PERIMETER

$P = a + b + c + d$

AREA

$A = \frac{1}{2}(a + b)h$

a and b are the two parallel sides.



Rhombus

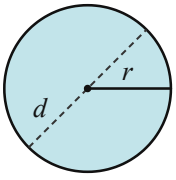
PERIMETER

$P = 4s$

AREA

$A = \frac{1}{2}d_1d_2$

The area formula also works for a kite.



Circle

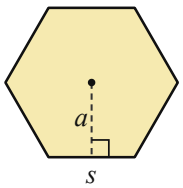
CIRCUMFERENCE

$C = 2\pi r = \pi d$

AREA

$A = \pi r^2$

The diameter is twice the radius: $d = 2r$.



Regular polygon

PERIMETER

$P = ns$

AREA

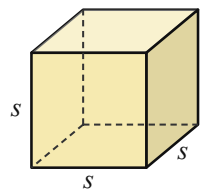
$A = \frac{1}{2}aP$

n equal sides of length s. The apothem a runs from the centre to the middle of a side.

Volume and surface area of solids

Surface area (SA) is the total area of every face of a solid, in square units. Dashed lines are edges hidden at the back.

Cube



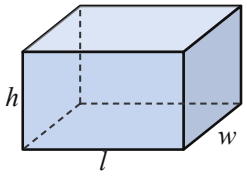
VOLUME

$V = s^3$

SURFACE AREA

$SA = 6s^2$

Rectangular prism



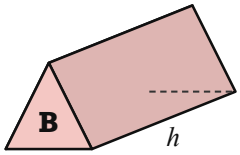
VOLUME

$V = lwh$

SURFACE AREA

$SA = 2(lw + lh + wh)$

Any prism



VOLUME

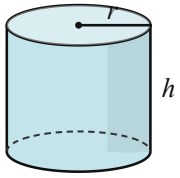
$V = Bh$

SURFACE AREA

$SA = 2B + Ph$

B is the area of the base, P its perimeter, and h the distance between the two bases.

Cylinder



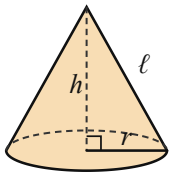
VOLUME

$V = \pi r^2 h$

SURFACE AREA

$SA = 2\pi r^2 + 2\pi rh$

Cone



VOLUME

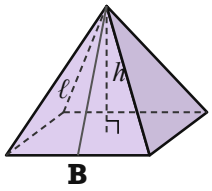
$V = \frac{1}{3}\pi r^2 h$

SURFACE AREA

$SA = \pi r^2 + \pi r\ell$

h is the height straight up; ℓ is the slant height along the side.

Pyramid



VOLUME

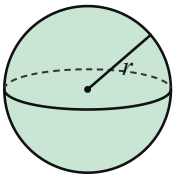
$V = \frac{1}{3}Bh$

SURFACE AREA

$SA = B + \frac{1}{2}P\ell$

The surface area formula is for a regular pyramid. P is the perimeter of the base.

Sphere



VOLUME

$V = \frac{4}{3}\pi r^3$

SURFACE AREA

$SA = 4\pi r^2$

Good to know

- $\pi \approx 3.14$ or $\frac{22}{7}$. Use the π key on a calculator for the most exact answer.
- **Pythagorean theorem:** in a right triangle with legs a, b and hypotenuse c, $a^2 + b^2 = c^2$. Use it to find a missing height or slant height.
- **Pointed solids** (cones and pyramids) hold $\frac{1}{3}$ of the matching prism or cylinder.
- **Scaling:** multiply every length by k and the area is multiplied by k^2, the volume by k^3.